Digital strategies for the social sciences – 7

Geometric Data Analysis

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INTRODUCTION

Questions about last class?

Geometric Data Analysis

INTRODUCTION

A Dimensionality reduction technique

Geometric Data Analysis

INTRODUCTION

A Dimensionality reduction technique

- Dimensionality Reduction: A procedure meant to reduce the number of variables under consideration, while preserving most of the information.

Geometric Data Analysis

INTRODUCTION

A Dimensionality reduction technique

• Dimensionality Reduction: A procedure meant to reduce the number of variables under consideration, while preserving most of the information.

• In practice: embedding of points in a large space into a smaller dimensional space (generally 2D, but can be more)

==> Underlying hypothesis: “low intrinsic dimensionality”

Geometric Data Analysis

INTRODUCTION

A Dimensionality reduction technique

- Dimensionality Reduction: A procedure meant to reduce the number of variables under consideration, while preserving most of the information.
- In practice: embedding of points in a large space into a smaller dimensional space (generally 2D, but can be more)

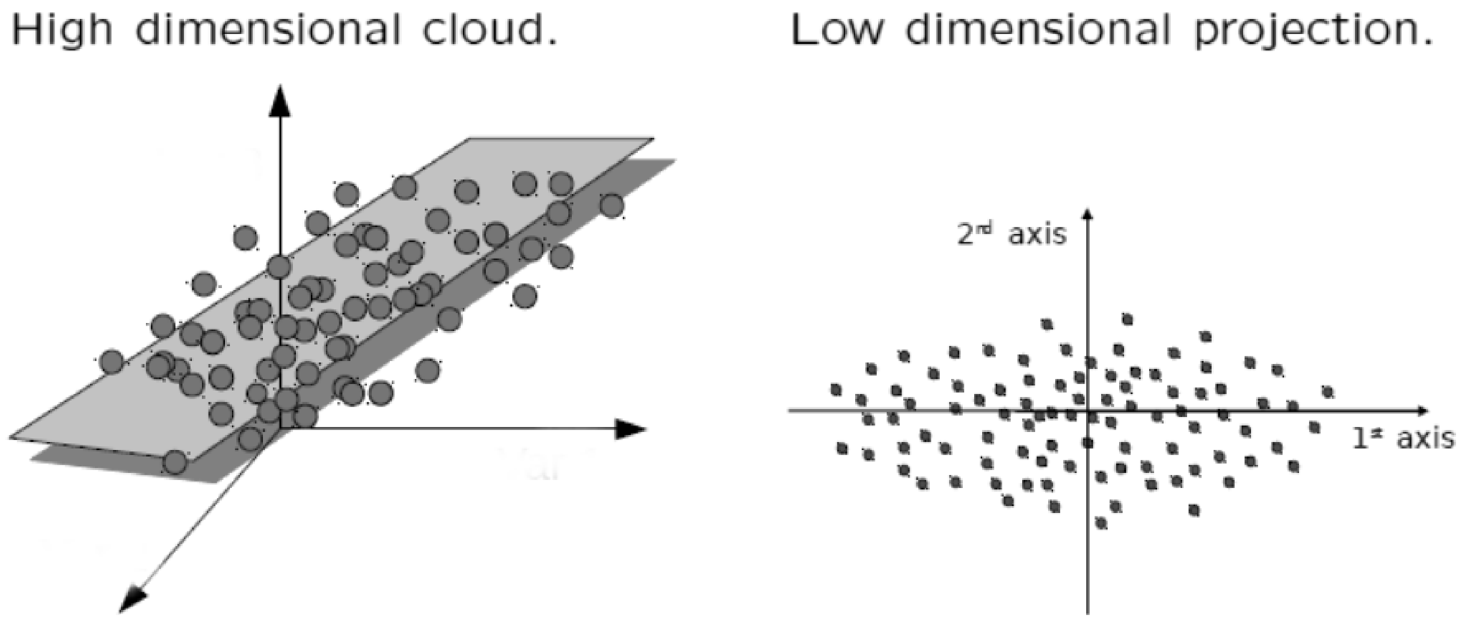
==> Underlying hypothesis: “low intrinsic dimensionality”

- Numerous techniques
    - Factorial analysis
    - PCA/MCA
    - MDS
    - ...and now machine learning.

Geometric Data Analysis

INTRODUCTION

A Dimensionality reduction technique



Geometric Data Analysis

INTRODUCTION

A Dimensionality reduction technique
History

• A classic technique, with a disputed genealogy (Pearson 1901, see also Goodman 1996)

• Debates about its uses for the social sciences

- EDA’s favorite (explore)
- Bourdieu’s use of it
- Feature extraction

Geometric Data Analysis

INTRODUCTION

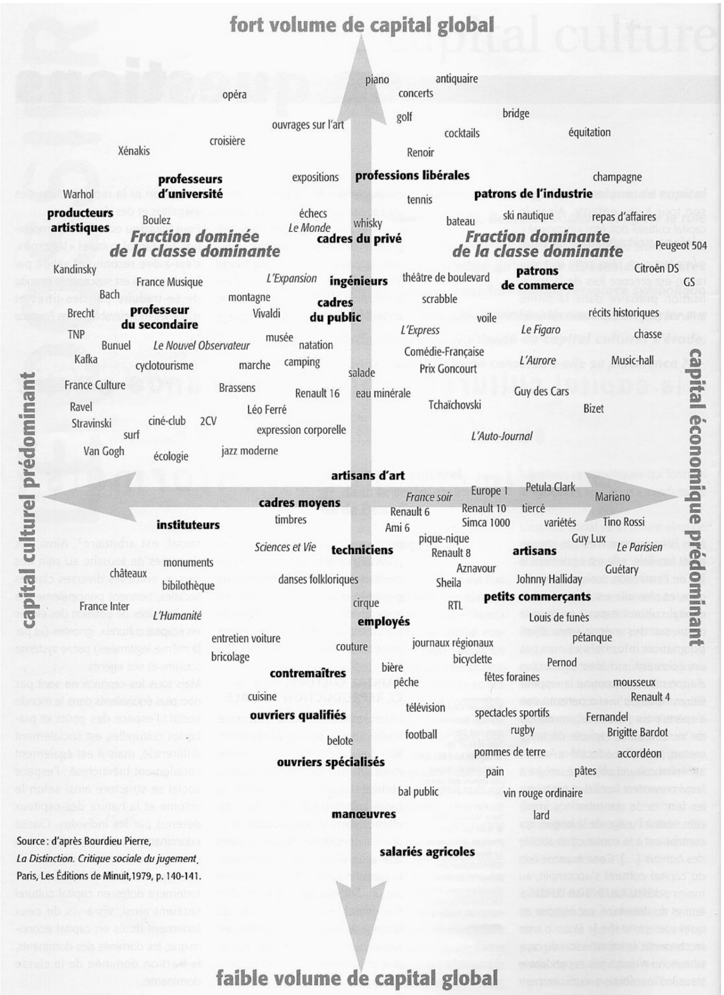
A Dimensionality reduction technique
History

• In the wake of Bourdieu, birth of a label: “Geometric Data Analysis”

Designates specific techniques (Principal Component Analysis; Multiple Correspondence Analysis, …) → Same family

- “Geometric”: measures of distance in space.

Geometric Data Analysis



Bourdieu, *Distinction*, 1979

“THIS IS NOT GDA” !
(a sociological parenthesis)

“The space of tastes and the social space” (made from concentrate of GDA)

Based on a study of cultural practices in 1970s France

Goal: finding people with similar practices and representing them on a 2D plane

From that to a theory of the social world

INTRODUCTION

Outline of the course

1. Principles
2. What does it do?
3. How to read a GDA
4. In practice: how to do it

Next session: advanced options and specifications

Geometric Data Analysis

PRINCIPLES OF GDA

- **Intuition**
  - ✓ Finding patterns between individuals based on the frequency of their practices (as measured by variables)

- **Relationship to other methods**
  - With respect to descriptive statistics
  - With respect to regression

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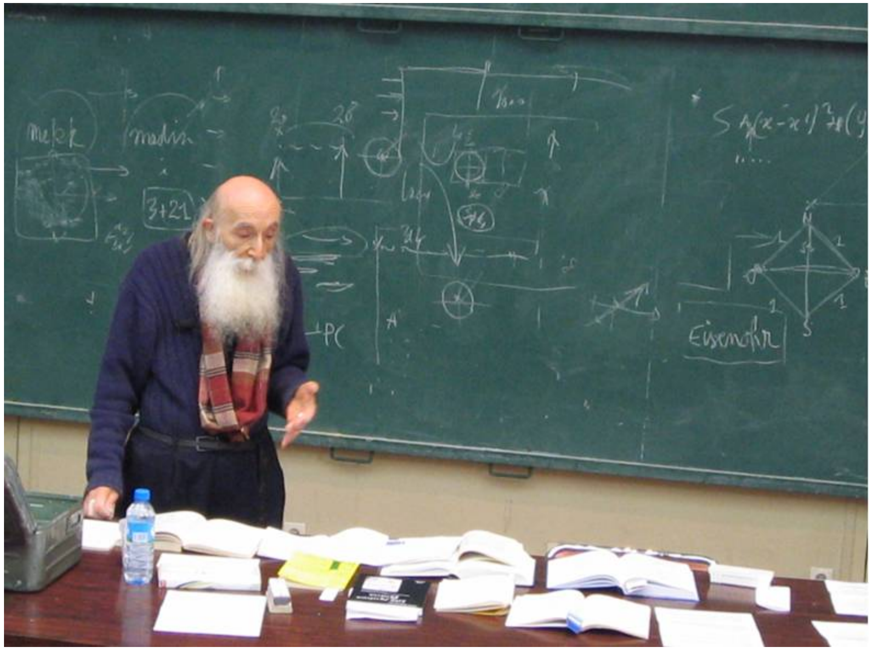
PRINCIPLES OF GDA

- **History**
  - ✓ There is a long an conflicted history about the method.
  - ✓ In France, it is often dated to French mathematician Jean-Paul Benzécri, who extended PCA to categorical variables, and developed many tools to perform dimensionality reduction.



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PRINCIPLES OF GDA

- **History**
  - ✓ This history is contested by some, who point to other ancestors (Pearson), or argue that the technique is similar to variance analysis performed in regression (see Goodman, 1996)
  - ✓ For a long time completely invisible in the US, but a rising trend in publications using it (Bourdieu + data avalanche)

Geometric Data Analysis

PRINCIPLES OF GDA

- General idea

	TV Public	TV Private	CSpan	Radio Public	Radio Private	TV Cable	Radio Cable	Internet
MP1	1	2	5	5	3	1	1	0
MP2	0	0	0	0	0	0	0	0
MP3	0	0	1	0	0	0	0	0
...								
MP576	0	0	1	1	0	0	0	0
MP577	15	12	12	16	19	9	12	10

Media invitations of MPs in France

Geometric Data Analysis

GDA: HOW TO READ IT

	TV Public	TV Private	CSpan	Radio Public	Radio Private	TV Cable	Radio Cable	Internet
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Media invitations of MPs in France

From this, you may want to know

- Which Mps have the same media practices ?
- Which Media invite the same Mps ?
- What is the overall structure of the data ?
- …

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PRINCIPLES OF GDA

- **General idea**
  - ✓ The goal is to find the vectors (often a 2D plane) that cut across it in a way to preserve most of its “structure” (in this case: variance)

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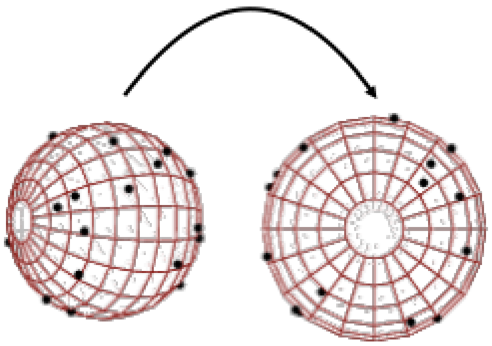
PRINCIPLES OF GDA

- **General idea**
  - ✓ The goal is to find the 2D plane that cuts across it in a way to preserve most of its structure (variance)



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PRINCIPLES OF GDA

- **General idea**
  - ✓ In both cases, you may consider your data table as an *n dimensional cloud of points.*
    - With continuous variables, n is the number of columns
    - With categorical variables, n is the (number of levels – the number of variables)
  - ✓ The goal is to find the 2D plane that cuts across it in a way to preserve most of its structure (variance)

(Detailed logic in the lab this afternoon – and in the video)

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PRINCIPLES OF GDA

Principle Component Analysis (PCA)

- On <u>continuous</u> variables
- Think about scaling (standardizing)
- Two main outputs

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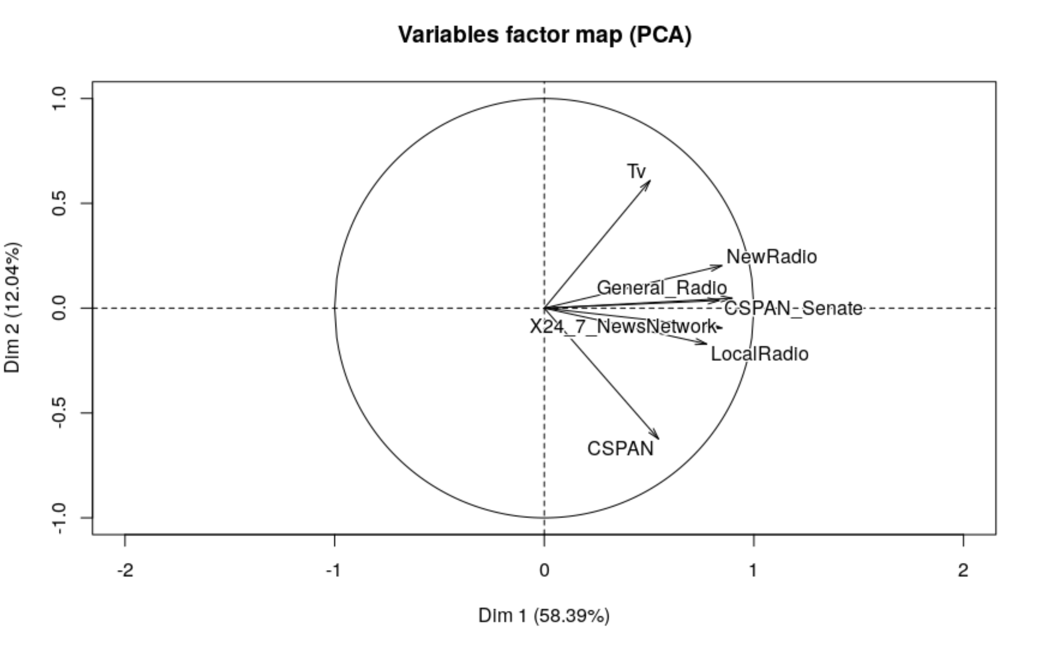
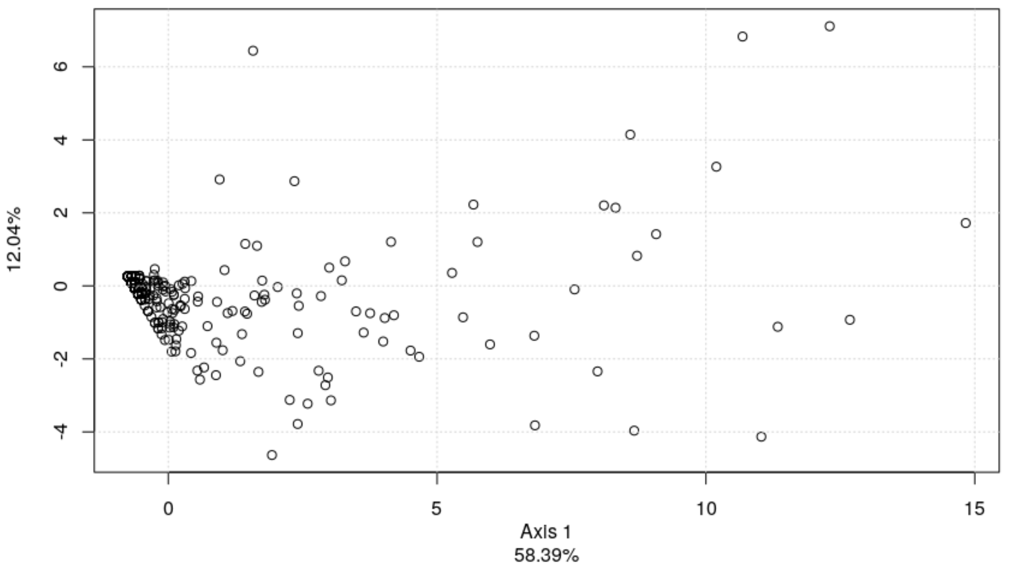
PRINCIPLES OF GDA

PCA

- Two main outputs





'Wheel' of correlations

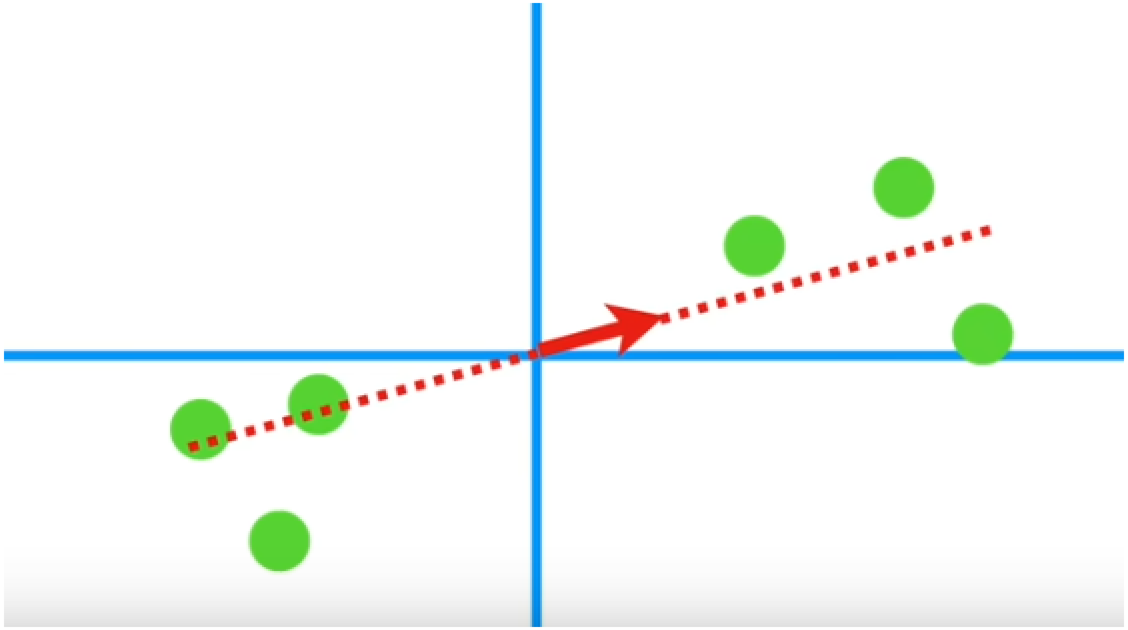
Cloud of points (individuals)

Geometric Data Analysis

GDA: WHAT DOES IT DO?

1. Creates an axis across the cloud of point

- Criterion: variance maximization
- Each initial variable contributes to the new axis
- Each individual/variable is positioned (coordinates) on the new axis



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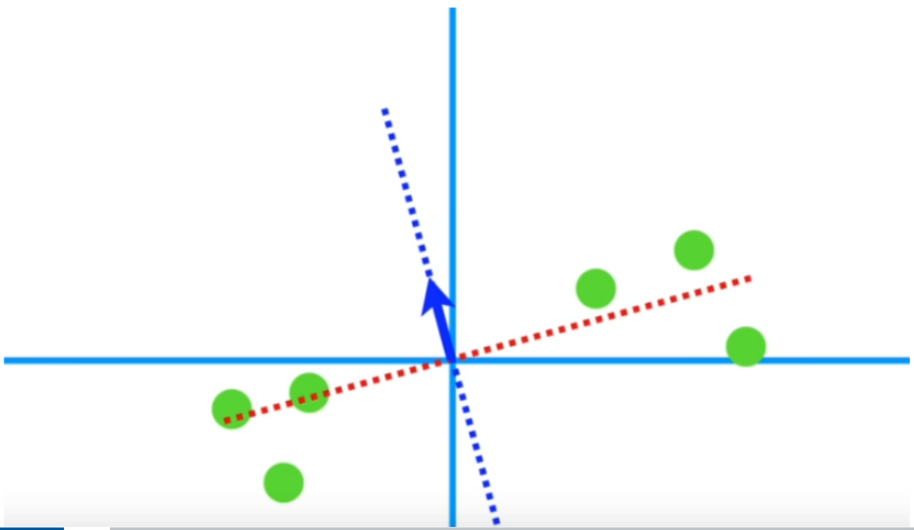
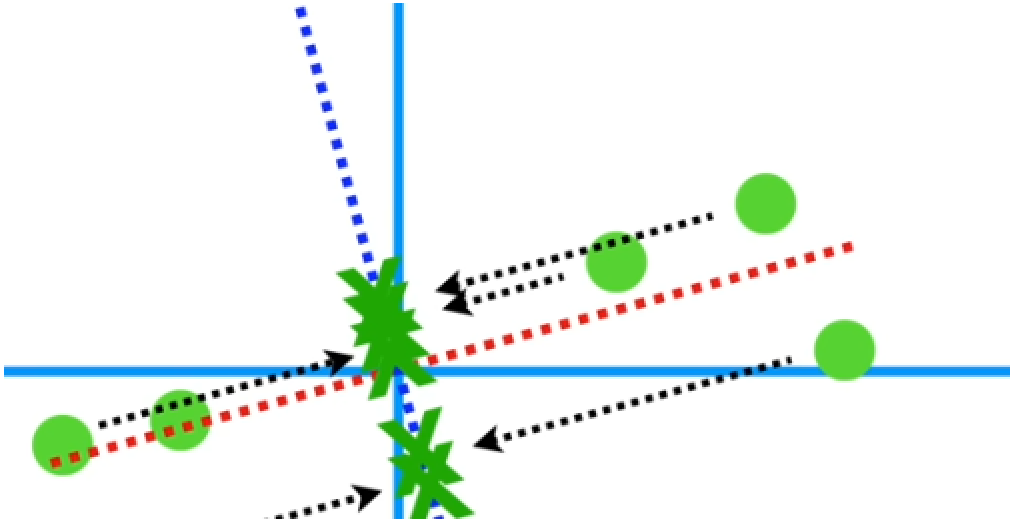
GDA: WHAT DOES IT DO?

2. Creates a 2nd axis across the cloud of point

- Criterion: variance maximization
- Each initial variable contributes to the new axis
- Each individual/variable is positioned (coordinates) on the new axis





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GDA: WHAT DOES IT DO?

3 and + Creates a 3rd axis, etc. etc.

If you have n initial variables, you can have as much as n-1 axes

More on that this afternoon

Geometric Data Analysis

GDA: HOW TO READ IT

	TV Public	TV Private	CSpan	Radio Public	Radio Private	TV Cable	Radio Cable	Internet
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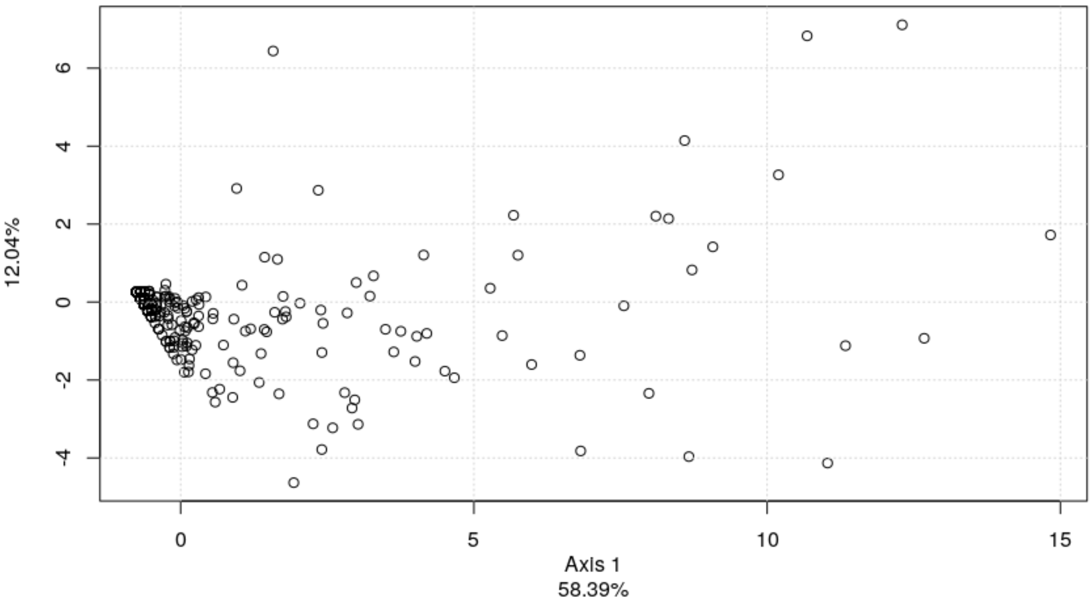
Media invitations of MPs in France

GDA: HOW TO READ IT

Let's take this step by step

Geometric Data Analysis

GDA: HOW TO READ IT



1. What is the quality of the data reduction we've realized?

- Measured by the eigenvalue (the variance of the axis)
- Here: 58 + 12
- Look at the cloud of points: good dispersion?

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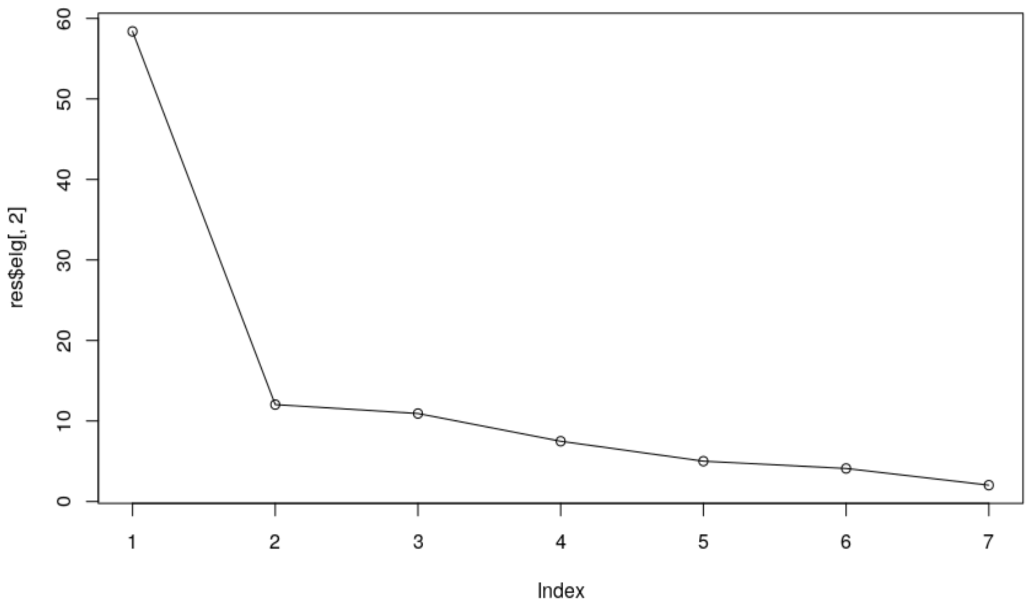
GDA: HOW TO READ IT



2. Do we need to look at another axis?

- You can (and sometimes) go beyond 2 axes

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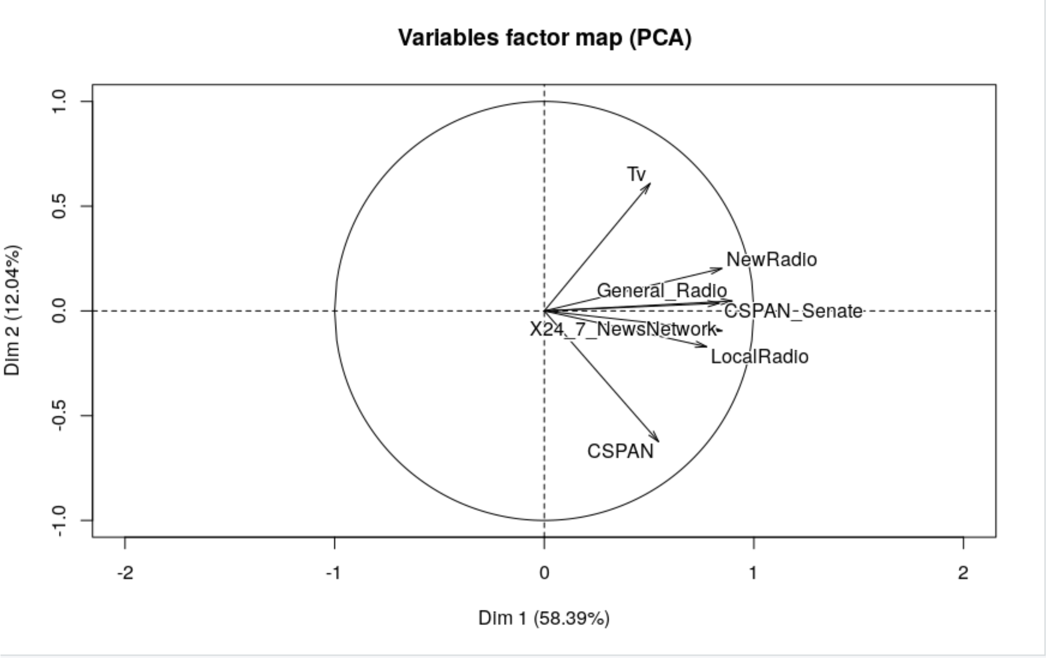
GDA: HOW TO READ IT



3. Which variables are 'opposed' on each axis?

Interpreting an axis amounts to finding out ***what is similar****, on the one hand, between all the elements figuring on the right of the origin and, on the other hand between all that is written on the left;* ***and expressing with conciseness and precision, the contrast*** *(or opposition)* ***between the two extremes****.*

Benzécri (1992, p. 405)

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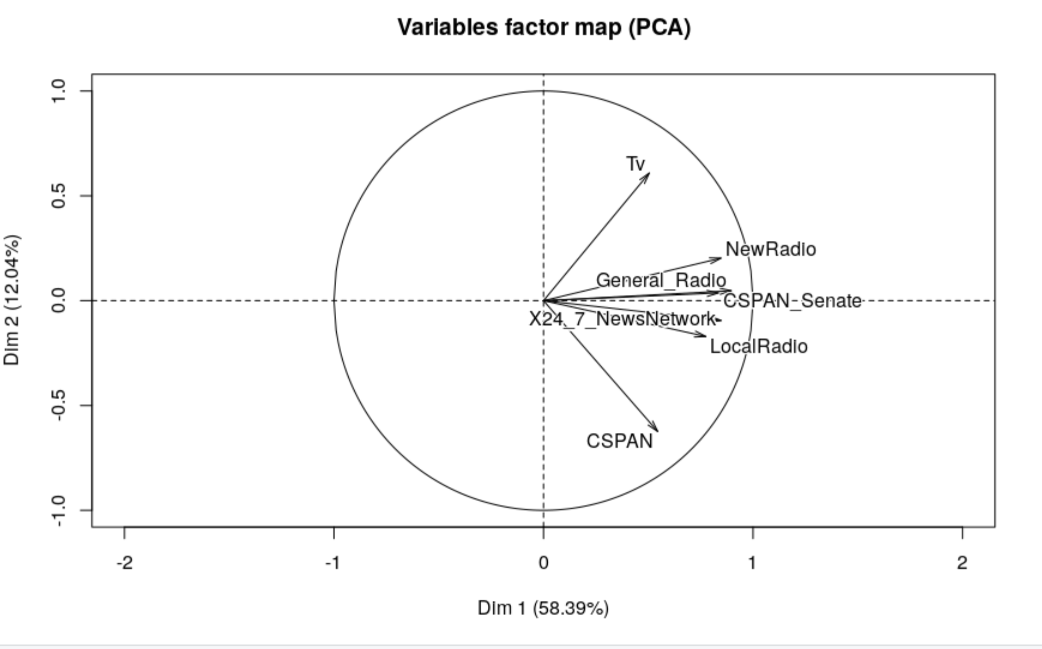
GDA: HOW TO READ IT



%

4. Provide an interpretation for the axes

- *Axis 1: All media*

→ ***Often invited***

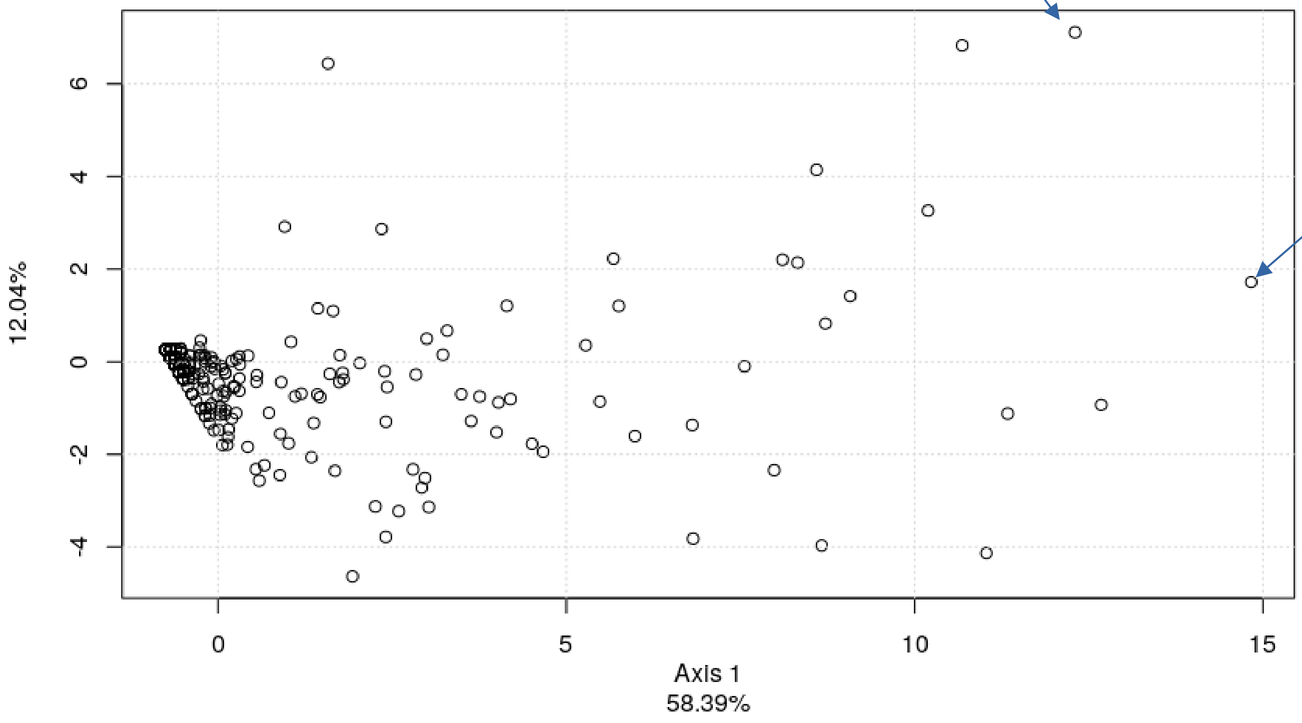
- *Axis 2: Opposition CSPAN vs. TV*

→ ***External vs internal invites***

Geometric Data Analysis

GDA: HOW TO READ IT

5. You can then look at where the individuals stand



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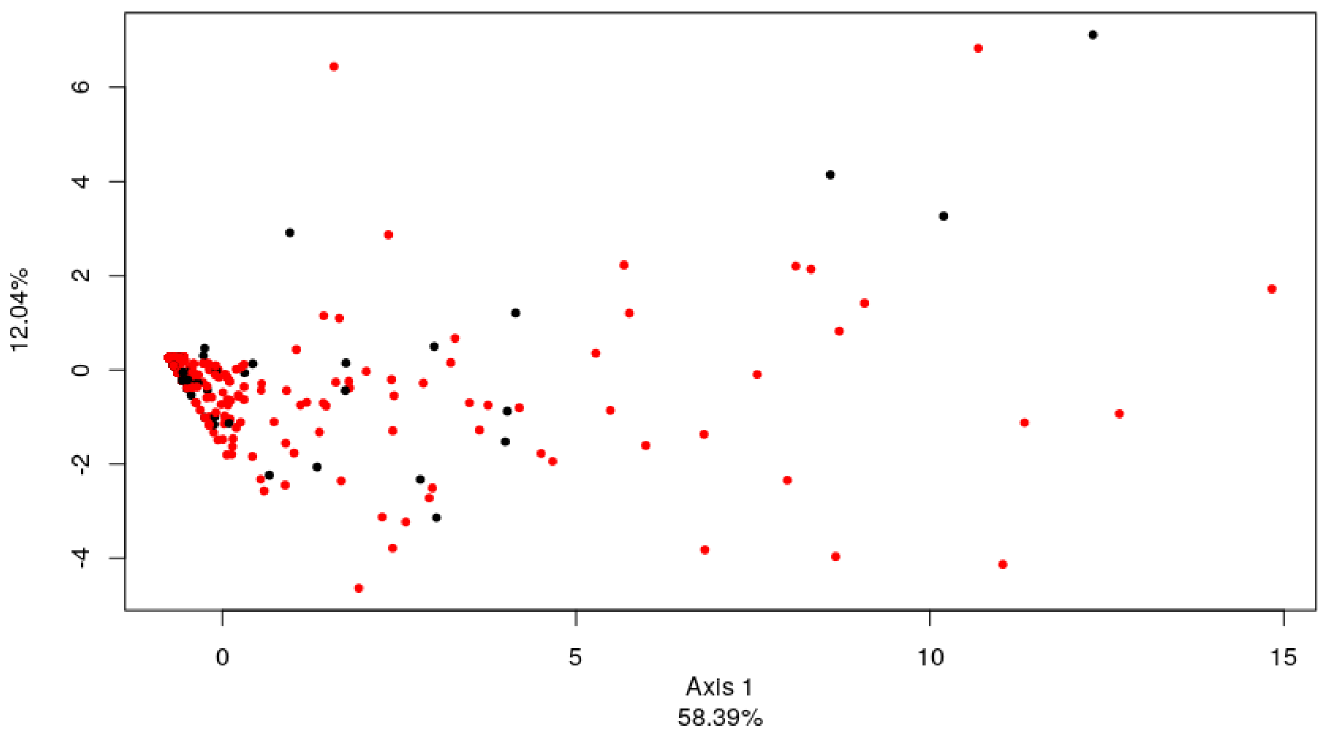
GDA: HOW TO READ IT

5b. You can color individuals by a given variable (gender here)



Geometric Data Analysis

IN PRACTICE

In R, several packages

The most comprehensive: FactoMineR (mind the case)

```
> PCA(dataset)
```

Geometric Data Analysis

IN PRACTICE

In R, several packages

The most comprehensive: FactoMineR (mind the case)

PCA(dataset)

Several options

- "Supplementary variables": Not used to compute the axes, but potentially informative
  - quali.sup → categorical variables
  - quanti.sup → continuous variables

- "Supplementary individuals": Not used to compute the axes, but potentially informative
  - ind.sup

Which line index ?

Which col. Index ?

PCA(dataset, ind.sup=256, quanti.sup=c(1:3))

Geometric Data Analysis

IN PRACTICE

In R, several packages

A very useful GUI: *explor()*

obj ← PCA(dataset)
#obj will store ***many*** useful information about the PCA (go look at it)

explor(*obj*)

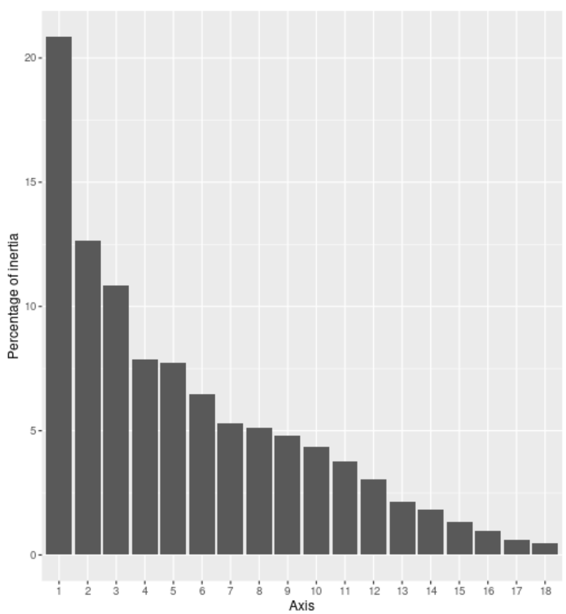
Geometric Data Analysis

IN PRACTICE

In R, several packages

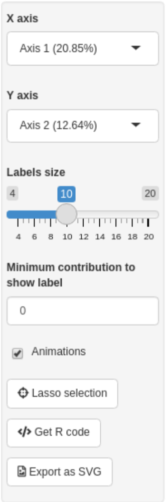
A very useful GUI: explor()
obj ← PCA(dataset)
explor(*obj*)

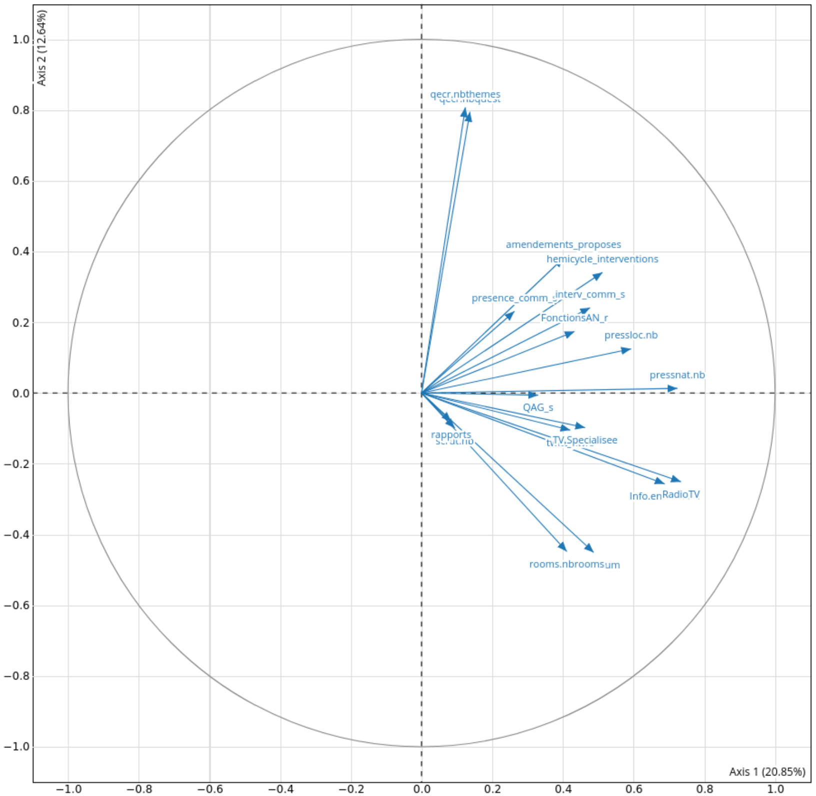
Eigenvalues histogram



Eigenvalues table

Axis	%	Cum.
1	20.8	
2	12.6	
3	10.8	
4	7.8	
5	7.7	
6	6.5	
7	5.3	
8	5.1	
9	4.8	
10	4.3	
11	3.8	
12	3.0	
13	2.2	
14	1.8	
15	1.3	
16	1.0	
17	0.6	
18	0.5	1



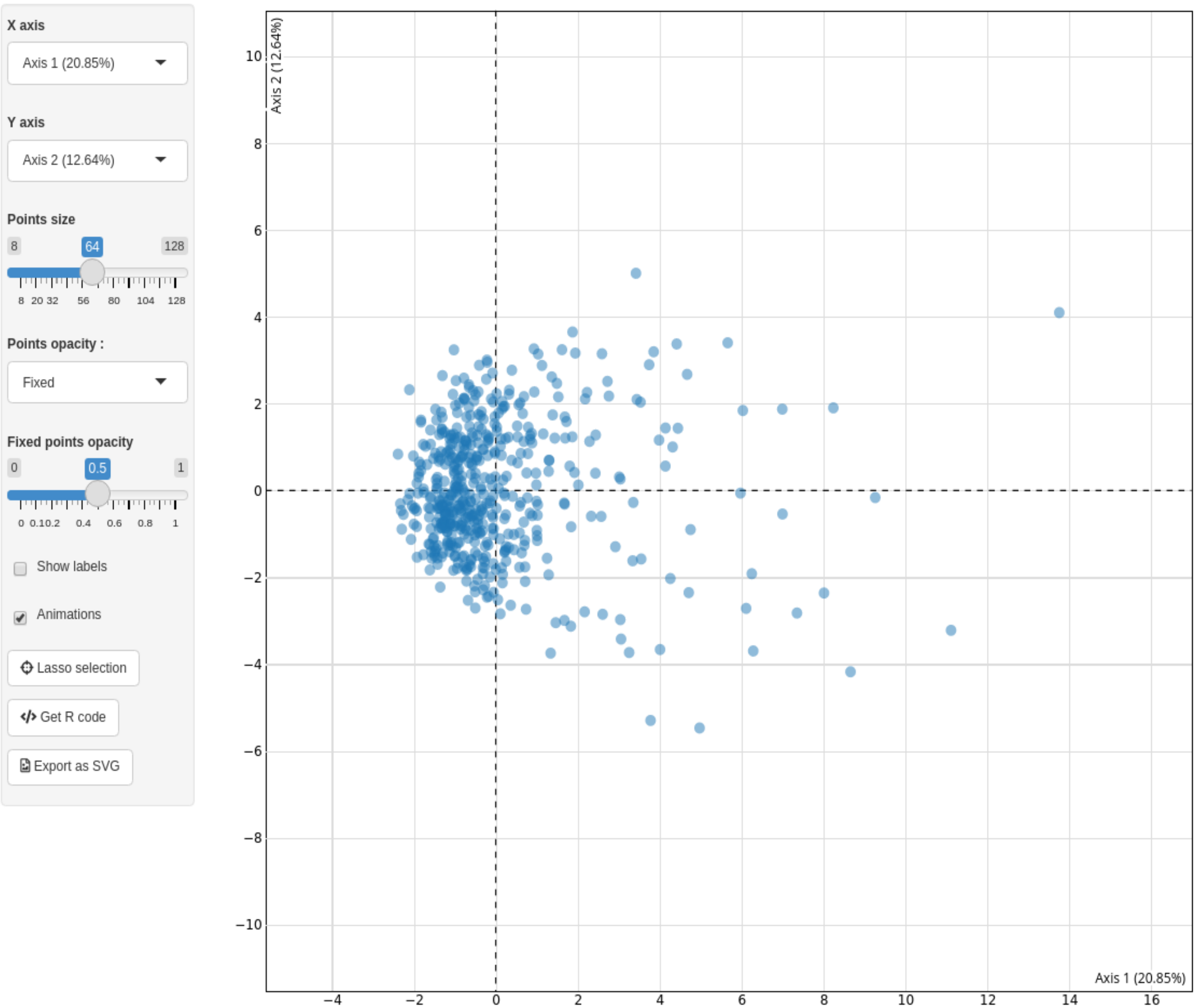


Geometric Data Analysis

IN PRACTICE

In R, several packages

A very useful GUI: explor()
obj ← PCA(dataset)
explor(*obj*)



Geometric Data Analysis

<u>This afternoon</u>

- The logic of PCA, in figures and in numbers
- How to in R

<u>Next time</u>

- On Categorical Variables: MCA
- Quality assessment
- Links to supplementary variables (age, gender ↔️ practices)
- Tips & Tricks

Introducing a data set

CASE STUDY

Valkompassen, 2014

- An investigation of the beliefs of voters, of politicians

- Carried out by SVT, every 4 years, with the help of political scientists
  - 45 questions on current topics, curated by political scientists
  - A series of socio-demographic information
  - In our case: MPs who responded (~245)

CASE STUDY

Example: Valkompassen, 2014
- An investigation of the beliefs of voters, of politicians

Questions:

- Can we sort candidates according on different value scales
- Is left/right enough to characterize SWE politics?
- How many dimensions structure Swedish politics?
- How cohesive are party in terms of values
- …

CASE STUDY

Example: Valkompassen, 2014

How to do this? GDA!

- Select your variables (why those?)
- Run your GDA
- Interpret it
- Comment

CASE STUDY

Example: Valkompassen, 2014